

ENGR 689 - Multimodal LLM Agents (Fall 2026): Quiz 2

Don't forget to write your name on the paper you submit!

Each question has only one correct option.

8 questions Duration: 25 Minutes

Q1 (1%) Which system most clearly illustrates an LLM agent rather than a standalone LLM?

- A. An LLM summarizes a document using only the text provided in its prompt.
- B. An LLM observes the current state of a web page, chooses an action such as clicking or typing, observes the resulting page, and then decides what to do next.
- C. An LLM generates eight independent answers and returns the most frequent one.
- D. An LLM is fine-tuned once on a collection of labeled text-classification examples.

Q2 (2%) Suppose an LLM is trained to answer questions by searching the web, but no reference search trajectories are provided. In one sampled trajectory, the LLM generates a search query, the search engine returns a passage, and the LLM then generates reasoning steps and a correct final answer. Under the outcome-based reinforcement learning approach discussed in class, which statement is correct?

- A. Every token in the trajectory, including the passage returned by the search engine, is treated as an LLM-generated token and reinforced.
- B. Only the final-answer tokens are used for training; the search query and reasoning steps generated by the LLM are ignored.
- C. The LLM-generated search query, reasoning steps, and final answer are part of the trainable trajectory, while the passage returned by the search engine is treated as an external observation and excluded from the model-probability computation.
- D. The model parameters cannot be updated because no reference search trajectory is available.

Q3 (2%) I am using Group Relative Policy Optimization (GRPO) to train an LLM to learn how to call a search engine as part of answering questions.

One of my training samples is:

Question: Who has won the Nobel Prize in Physics twice?

Answer: John Bardeen

I then let the LLM attempt this question four times, with the following trajectories:

Trajectory 1: search → reasoning → John Bardeen

Trajectory 2: search → reasoning → search → reasoning → John Bardeen

Trajectory 3: search → reasoning → Barry Sharpless

Trajectory 4: search → reasoning → search → reasoning → Frederick Sanger

I define the reward for each trajectory as follows:

$$\text{reward} = 5 \times \mathbf{1}[\text{output} = \text{John Bardeen}] - 1 \times \# \text{search}.$$

In other words, if the final answer is correct, it receives +5 points. Additionally, to encourage efficient use of the search engine, each call to the search engine incurs a penalty of −1 point.

Under the GRPO algorithm, which trajectory (or trajectories) would be considered “good”?

- A. Trajectory 1
- B. Trajectory 1 and Trajectory 2
- C. Trajectory 1 and Trajectory 3
- D. Trajectory 1, Trajectory 2, and Trajectory 3

Q4 (1%) Which sequence most directly represents a ReAct-style agent trajectory?

- A. Reasoning → Final answer
- B. Action → Observation → Action → Observation
- C. Reasoning → Action → Observation → Reasoning → Action → Observation
- D. Generate several complete reasoning paths → Select the most frequent final answer

- Q5 (1%)** An agent produces an incorrect plan and receives detailed textual feedback explaining the problem. The model then writes a reflection, includes that reflection in the prompt for its next attempt, and tries again without updating its parameters. Which method does this most directly illustrate?
- A. Supervised fine-tuning
 - B. Standard reinforcement learning with numerical rewards
 - C. Verbal reinforcement learning
 - D. Group Relative Policy Optimization
- Q6 (1%)** Which design best addresses the context-window limitations of an agent performing a long-horizon task while also capturing a key difference between agent memory and an ordinary RAG corpus?
- A. Include the agent's entire interaction history in every prompt, regardless of its length.
 - B. Store records from previous interactions, retrieve the most recent and relevant information when needed, and allow the agent to write to and update its own memory.
 - C. Retrieve information only from a fixed external corpus written by others and never store the agent's own experiences.
 - D. Fine-tune the model parameters after every action so that no previous records need to be stored.
- Q7 (1%)** A system must rank more than one million documents. In the setup discussed in class, lexical matching is efficient, while a BERT-based ranking model is more effective but computationally expensive. Which pipeline best balances efficiency and effectiveness?
- A. Use BERT to rank all documents and then apply lexical matching to the highest-ranked documents.
 - B. Use lexical matching to retrieve a small set of candidate documents and then use BERT to rerank those candidates.
 - C. Randomly sample a small set of documents and then rank them using BERT.
 - D. Use BERT to encode only the query and rank documents without examining their content.
- Q8 (1%)** In Toolformer-style training-data construction, candidate API calls are inserted into ordinary text and then filtered. Which criterion is most important for deciding whether an inserted API call should be retained?
- A. The API call is topically related to some part of the preceding text.
 - B. The API call uses the most sophisticated tool available to the model.
 - C. The API response increases the model's probability of generating the subsequent text.
 - D. The API call is inserted near the beginning of the sentence.